Week 2 Discussion

CSE 11 Winter 2026

Open Discussion Section 2 Interactive Groupwork on Gradescope

Logistics

- We're using the Autograder queue
- Start your Assignments early!
- Office hours are for getting you unstuck and/or addressing any doubts you have - not for paired programming help

How to use Piazza

Use Piazza to…

- Ask **course-content questions** that other students may also have
- Ask for help understanding **Java concepts, syntax, or error messages**
- Clarify **assignments, worksheets, deadlines, and course policies**
- **Search before posting** — your question may already be answered!
- Answer classmates' questions when you can

When posting

- Use a **descriptive title**: *"Why does this String comparison return false?"*
- Include the **relevant code/error message**
- Explain **what you tried** and what you expected
- Post **privately** if your question contains assignment-specific code that could reveal a solution

Remember: Piazza is for getting help, **not posting complete solutions**. Keep questions respectful and give instructors/TAs time to respond.

Boolean Expressions

`true`

`false`

`2 > 5` → **false**

```
String s = "hello";
```
`s == "hello"` → **false**

`s.equals("hello")` → **true**

`&&` → **AND**

`||` → **OR**

`true && false` → **false**

`true || false` → **true**

Should the student get all of their social learning credit for Week 1?

```
(student.isEnrolled() || student.isWaitlisted())
    && student.attendedLectures(1)
    && student.attendedDiscussion(1) && student.read(1)
```

`if-else` statements

```
if (boolean) {
    ...
}
```

```
if (boolean) {
    ...
} else {
    ...
}
```

```
if (boolean) {
    ...
} else if (boolean) {
    ...
} else if (boolean) {
    ...
} else {
    ...
}
```

PA1 Overview



- Analog clock
  - **Given**: time of day
  - **Output**: 5 .png files that draw a clock

Q: Write a program that draws a rectangle and rotates it.
A: Live coding session…

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```
1   void main() {
2       Pict rectangle = new Rect(10, 70, "red");
3       ImageIO_.save(rectangle, "rectangle.png");
4       Pict turnt = new Rotate(rectangle, 45);
5       ImageIO_.save(turnt, "rotated-rectangle.png");
6   }
```

Q: How would you rewrite `turnt` using syntax that is nested? Answer in the box provided on the Gradescope assignment.

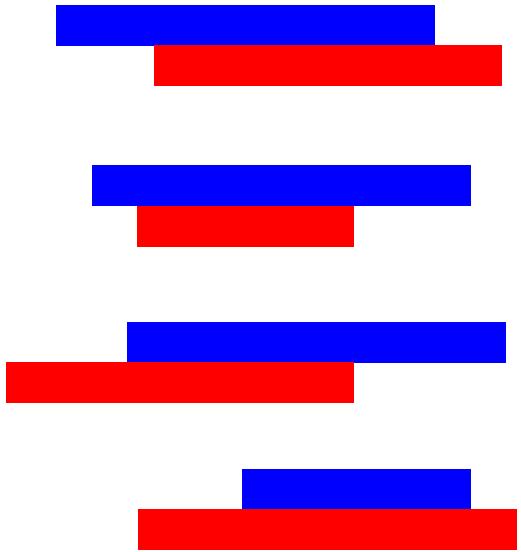
PA1 Overview

- Overlap
  a. Write a program with a `record` to represent a lecture session: **day of the week**, **time**, and **duration**
  b. Write a method that takes another instance of the record and reports whether the two times overlap
  c. Hint: think of all the ways two lectures could overlap

Q: What do the following evaluate to? Discuss with a partner (2 min) (`true`, `false`, errors out)

```
String day = “Wednesday”;
int nBurritos = 6;
```

```
1) day.equals(“Wednesday”);
2) nBurritos == 6;
3) true == true;
4) day != nBurritos;
```

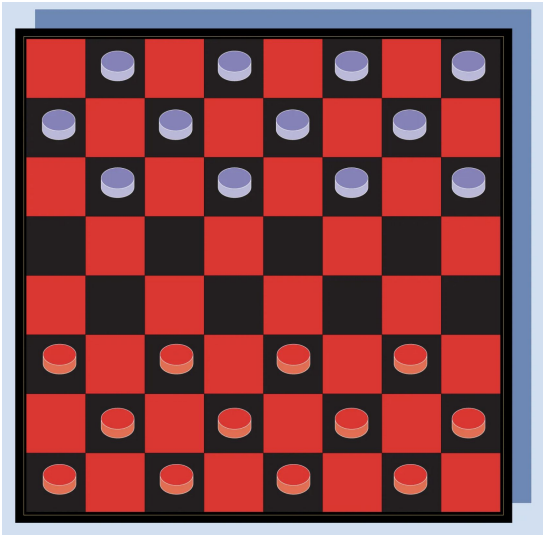


PA1 Overview

- Checkerboard
  a. Write a definition of a checkerboard image (8x8, alternating square colors). Do it without using loops.

You could explicitly draw each of the 64 squares with using above and beside. Is this the most efficient strategy?

Q: Draw some pink “polka dots”
A: Live coding…



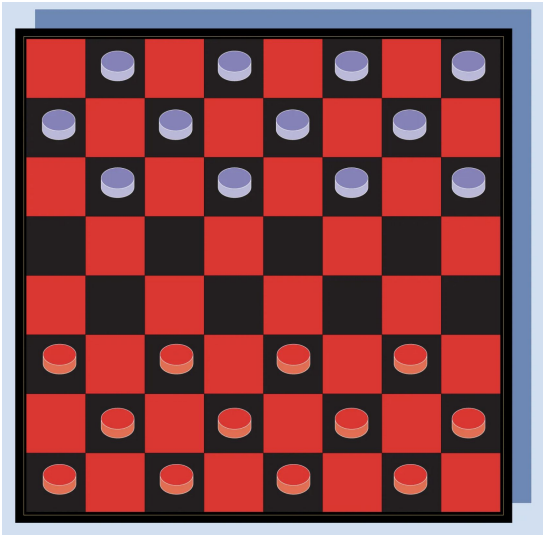
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PolkaDots.java

```
1   void main(){
2       Pict dot1 = new Circle(20, "pink");
3       Pict dot2 = new Circle(20, "transparent");
4
5       Pict dots1 = dot1.beside(dot2);
6       Pict dots2 = dot2.beside(dot1);
7
8       Pict dots = dots1.beside(dots1).above(dots2.beside(dots2));
9
10      ImageIO_.save(dots, "polkadots.png");
11  }
```



PA1 Overview

- Valid Name: design a method and analyze an AI implementation
  a. Imagine you have a job at a company related to account sign up.
     Design (don't implement yet) a method that takes a String and reports if it is likely a valid last name.
  b. AI-generated implementation

Design Recipe

1. Define the data and write examples to make the representation clear
2. Identify the method you want to write
3. Write the method header and examples together to figure out what it's supposed to do; placeholder values to make it run
4. Implement the method body and check it against the examples.

Exercise

Design a program to keep track of a library book. A book has a title, an author, and a number of days it can be borrowed for free. If the book is kept longer than the free period, there is a daily late fee. Write a method that calculates the total late fee after a given number of days.

1. Define the data and write examples to make the representation clear
2. Identify the method you want to write
3. Write the method header and examples together to figure out what it’s supposed to do; placeholder values to make it run
4. Implement the method body and check it against the examples.

Agenda

- PA1 Overview
  -

Thank You!
Any Questions?